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Roll No

EC-705 (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
VLSI Design
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. If 1000 devices on a chip must have specific parameter within the specified design process window, determine the soft yield if the process has been characterized by a complexity index of :
 - i) $C_p = 0.5$
 - ii) $C_p = 1.0$
 - iii) $C_p = 1.5$
 - iv) $C_p = 2.0$
2. Explain IC production process and what are the methods of testing in detail.
3. Discuss the design and applications of bit serial processing elements.
4. Explain the operation of N-channel Enhancement MOSFET with different values of gate voltage.
5. Discuss the various design rules of CMOS. What are the significance of these rules in software system?
6. What do you mean by systolic arrays? Draw any circuit which explains the effective results of these type of arrays.
7. Discuss the principles of Gausi-static register cell.

OR

Discuss about the bipolar process in detail.

8. Write short notes: (any two)
 - a) Latch up
 - b) Micro coded controllers
 - c) BJT Noise Model
 - d) Device Modelling
 - e) Twin Tub Process
